Lab 9 - Object Properties and Layers

Course: WSQ Technical Drawing with AutoCAD (TGS-2021005538) | Topic 04: Organizing Objects

Objective: LO3/LO4 - organise a drawing with layers and properties (A4, K4)

Goal

Create a standards-based layer scheme in lab-table-lesson2.dwg, then audit the supplied office-plan-ground-floor.dwg to see how a larger architectural drawing separates systems.

You'll produce

A layered table drawing where every object is ByLayer, plus a five-layer audit of the office plan covering naming, colour, linetype, lineweight and plot state.

Commands / tools: LAYER (LA), Properties palette (PR), Layer Walk, MATCHPROP, freeze/thaw, lock

Files in this lab

- lab-table-lesson2.dwg - AutoCAD drawing file
- office-plan-ground-floor.dwg - AutoCAD drawing file

Step-by-step

1. Open lab-table-lesson2.dwg and inspect current object properties

```
PR (Properties palette)
```

2. Create layers: A-OUTLINE, A-HIDDEN, A-CENTRE, A-DIMS, A-TEXT with colours and linetypes

```
LA → New Layer → colour/linetype
```

3. Load the HIDDEN and CENTER linetypes and set lineweights per layer

```
LA → Linetype → Load → HIDDEN, CENTER
```

4. Select objects by type and move them to their layers; copy formatting with MATCHPROP

```
select → Layer drop-down  |  MA
```

5. Practise layer states: turn off, freeze/thaw, lock and Layer Walk through the scheme

```
LAYWALK | LAYFRZ | LAYLCK
```

6. Set every object's colour, linetype and lineweight to ByLayer and save

```
SETBYLAYER → SAVEAS table-layers.dwg
```

7. Open office-plan-ground-floor.dwg and audit five functional layers against the same rules

```
OPEN office-plan-ground-floor.dwg → LA → record name/colour/linetype/lineweight/plot
```

Check your work

Turning each layer off hides exactly its objects, everything in table-layers.dwg reports ByLayer, and the office-plan audit identifies five layers with their intended functions.